Advanced Biometric Systems and Security (AI461)

Assignment 1
User Authentication using Biometric Features

The objective of this assignment is to let you gain some hands-on experience in the performance evaluation of a typical biometric system. You are provided with a **data file** that has biometric feature sets from each of the 100 users. Each user has provided 10 image samples: 5 during enrollment and 5 6 months after enrollment. The feature vectors from each of these 1000 image samples are stored in the data file. The system has acquired 144 features from each of the biometric images and generated a representative feature vector.

You have to use the first five image samples as training features and the remaining five as the test set to report test results.

The following performance curves/scores on the test data using (i) Euclidean distance and (ii) cosine similarity as the matching distance are to be computed.

A. Genuine Distribution
B. Imposter Distribution
C. FAR Plot
D. FRR Plot
E. Receiver Operating Characteristics
F. Equal Error Rate and Decidability Index

Which matching distance criteria, in your opinion, perform the best?

Points Distribution [Maximum 50]

1. Implementation/Results from A and B @ Max 20 Points
2. Implementation/Results from C and D @ Max 5 Points

3. Implementation/Results from E @ Max 15 Points

4. Results from F @ Max 10 Points

5. Clarity, Formatting and General Layout ® Up to -3 if not followed

6. Originality ® Both copier and originator get ZERO

7. Late Policy ® 20% reduction, allows ONLY one day delay.

Submission: README file with
Explanations about your implementation that may be useful for the grader

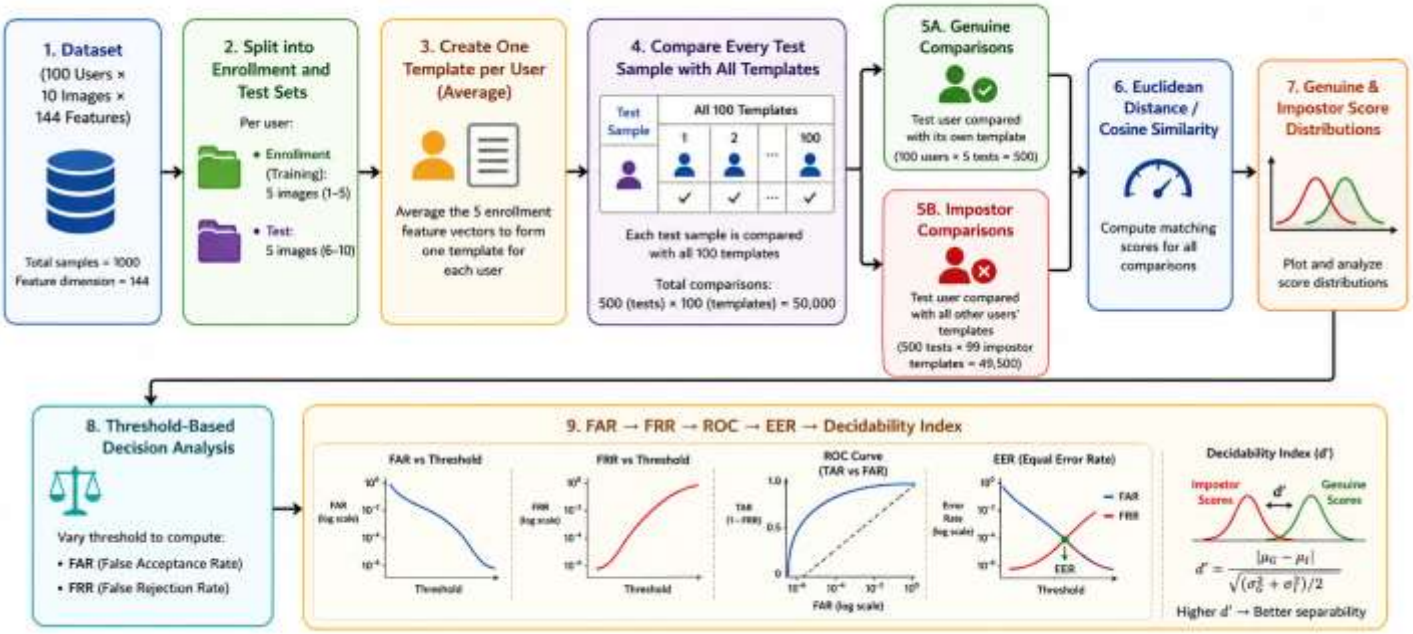
You will be called for a short demo at a convenient time/date. Please make sure that you upload the assignment to GitHub.

Deadline: The submission deadline for Assignment # 1 is 15 August 2026 [23:59 Hrs].

Questions

If you have any further questions relating to this assignment, you may contact the TA, Ms Priyanka Sharma

Inspired by: Prof Ajay Kumar, the Hong Kong Polytechnic University


1. Dataset
(100 Users × 10 Images × 144 Features)
Total samples = 1000
Feature dimension = 144
2. Split into Enrollment and Test Sets
Per user:
• Enrollment (Training): 5 images (1–5)
• Test: 5 images (6–10)
3. Create One Template per User (Average)
Average the 5 enrollment feature vectors to form one template for each user
4. Compare Every Test Sample with All Templates
Test Sample
All 100 Templates
1
2
100
Each test sample is compared with all 100 templates
Total comparisons:
500 (tests) × 100 (templates) = 50,000
5A. Genuine Comparisons
Test user compared with its own template
(100 users × 5 tests = 500)
5B. Impostor Comparisons
Test user compared with all other users' templates
(500 tests × 99 impostor templates = 49,500)
6. Euclidean Distance / Cosine Similarity
Compute matching scores for all comparisons
7. Genuine & Impostor Score Distributions
Plot and analyze score distributions
8. Threshold-Based Decision Analysis
Vary threshold to compute:
• FAR (False Acceptance Rate)
• FRR (False Rejection Rate)
9. FAR → FRR → ROC → EER → Decidability Index
FAR vs Threshold
FRR vs Threshold
ROC Curve (TAR vs FAR)
EER (Equal Error Rate)
FAR
FRR
EER
Threshold
FAR (log scale)
Decidability Index (d')
Impostor Scores
Genuine Scores
Higher d' → Better separability